

CarbonLedger

Software Requirements Specification

Calculation, factor versioning, uncertainty, period close and late-data rules

CL-SRS-001 · VERSION 1.0 · DRAFT FOR REVIEW

9

plants

3

emission scopes

4

data quality tiers

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Project: Emissions Accounting for a Multi-Plant Manufacturer

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Summary & Contents

Purpose of this document

Defines what CarbonLedger shall do so that every reported emissions figure is calculated correctly, reproducible from the same inputs and factors two years later, and disclosed with an honest uncertainty.

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Document control

Revision history

Version	Date	Description	Author (role)
1.0	03/10/2026	Initial release for review	Sole Engineer

Approvals

Role	Signature	Date
Sponsor (Manufacturer, Head of Sustainability)		
Reporting Lead (Manufacturer)		
Sole Engineer (all project roles)		
Reviewer (Faculty)		

Related documents

- CL-SRS-001 Software Requirements Specification (v1.0) · this document
- CL-DTT-001 Decision Table and Test Set (v1.0)
- CL-DSG-001 Design Pack (v1.0)
- CL-CMP-001 Configuration Management Plan (v1.0)
- CL-DQP-001 Data Quality Plan (v1.0)
- CL-PPL-001 Project Plan (v1.0)

Notice. This document is controlled; printed copies are uncontrolled. Figures stated as project data (data quality tiers of ±2%, ±8% and ±25%, nine plants, three scopes) come from the case study. Numbers in worked examples and test fixtures are synthetic, are labelled as such, and are not emission factors published by any authority.

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines what CarbonLedger shall do. CarbonLedger is a system that calculates and reports greenhouse gas emissions for a manufacturer with **nine plants** across **three scopes**, to a customer who has made the report a condition of a supply contract. The calculation itself is simple multiplication. The hard requirement is **auditability**: a figure reported today must be reproducible two years later, from the same inputs, with the emission factors that were valid at the time, and the system must say honestly how uncertain the figure is.

This SRS specifies calculation, emission factor versioning, uncertainty disclosure, period close and late-data rules. It is the parent of the Decision Table and Test Set (CL-DTT-001), Design Pack (CL-DSG-001), Configuration Management Plan (CL-CMP-001), Data Quality Plan (CL-DQP-001) and Project Plan (CL-PPL-001).

1.2 Scope

In scope (version 1): capture of activity data from electricity bills, fuel purchase records, logistics invoices and supplier declarations; Scope 1, 2 and 3 calculation; temporal emission factor storage; uncertainty by data quality tier; period close; late data handling; report generation and byte-level verification; audit trail.

Out of scope (version 1), stated so nobody assumes otherwise:

- Scope 2 market-based method (only one method, location-based, is implemented).
- Uncertainty of the emission factors themselves (see FR-UNC-08 for how this is disclosed).
- Biogenic emissions, offsets and removals.
- Automatic reading of bills by OCR, and live meter integration (data is entered or uploaded as CSV).
- Forecasting and target setting.

1.3 Definitions

Term	Meaning
Scope 1 / 2 / 3	Scope 1: direct emissions from fuel burned at the plants. Scope 2: emissions from purchased electricity. Scope 3: other indirect emissions, here inbound and outbound logistics and purchased goods covered by supplier declarations.
Activity record	One quantity of an activity (kWh, litres, tonne-km) for one plant, with a date interval, a data quality tier and evidence.
Emission factor	Published kgCO ₂ e per unit of activity, valid for a stated date interval.
Tier	Data quality class of a record: metered (±2%), invoiced (±8%), estimated (±25%), or missing (no value received).
Reporting period	A contiguous range of whole calendar months set by the customer contract. A period is OPEN, CLOSING or CLOSED.
Closed at	The instant a period is closed. It is the technical data cut-off used for reproduction.

1. Introduction

Term	Meaning
Late record	A record whose activity date lies in a period that is already CLOSED when the record is stored.
Reference release	An immutable, sealed bundle of emission factors, unit conversions, tier percentages and rounding policy, with a SHA-256 manifest hash.
Calc release	A frozen version of the calculation logic (module such as <code>calc.v1</code>).
Pair	A calc release together with a reference release, recorded as one baseline (Pair ID such as CL-2024.1).
Canonical report	The report serialised to a defined byte sequence (section 3.8). Its SHA-256 is the report hash.
CRD	Contract Reporting Date: the date by which the customer requires the report. Its value comes from the contract and is configuration, not hard-coded.

1.4 References

Case Study No. 38 statement; GHG Protocol corporate and Scope 3 standards (for scope definitions and the principle of reporting uncertainty); the emission factor publication of the publishing authority (name and edition recorded in each reference release). Project website: <https://example.com/carbonledger>.

2 Overall Description

2.1 Product perspective

CarbonLedger is a new, standalone web application used internally by the manufacturer. It replaces spreadsheets. Inputs arrive from four document types and each maps to a scope and a usual tier:

Input document	Typical scope	Typical tier	Notes
Electricity bill	2	invoiced (metered when a plant sub-meter reading is held)	Billing interval may cross month or period boundary.
Fuel purchase record	1	invoiced (metered when a flow meter is read)	Quantity in litres or kg.
Logistics invoice	3	invoiced; estimated when distance is assumed	Quantity in tonne-km.
Supplier declaration	3	estimated (unverified)	Reliability varies; never upgrades a tier by itself.

2.2 User classes

Role	Responsibility in the system
Plant Data Officer	Enters or uploads records and evidence for their own plant.
Plant Manager	Confirms the plant's data before close.

2. Overall Description

Role	Responsibility in the system
Reporting Lead	Starts and approves period close, issues reports, decides on restatement.
Reference Data Steward	Loads and seals reference releases from the authority's publication.
Auditor (internal or customer)	Read-only access, drill-down and the verify function.
Administrator	Accounts and backups. Cannot edit data or factors.

2.3 Operating environment and technology

Technology is deliberately plain so that one engineer can build, test and hand it over:

Layer	Choice	Justification
Language	Python 3.12	<code>decimal</code> module gives exact, deterministic decimal arithmetic; no binary floating point is needed.
Web	FastAPI with server-rendered Jinja2 pages	No JavaScript framework to version; fewer moving parts to reproduce later.
Database	PostgreSQL 16	Range types and exclusion constraints enforce non-overlapping factor validity; triggers enforce immutability.
Reports	Canonical JSON (hashed), PDF and CSV renderings (derived)	PDF layout is not byte-stable, so the hash is taken over the JSON, which is.
Source control	Git, one repository for code and reference data	One tag versions both together (see CL-CMP-001).
Tests and CI	pytest in a pinned container, run in CI on every commit	Same environment every time.
Deployment	Docker Compose on one server, daily database backup	Nine plants and low data volume do not justify more.

2.4 Constraints and assumptions

- The customer contract fixes the CRD and the period length (assumption A1: the values are supplied before design freeze).
- The authority publishes factors annually, and may issue corrections later (assumption A2).
- Plants can provide data as bills, records or CSV within the cycle calendar in CL-PPL-001 (A3).
- One engineer delivers the project (A4); automation gates replace a second reviewer where independence is needed.
- Data volume is small (nine plants, monthly records), so performance is a design target, not a risk driver (A5).

3. Functional Requirements

Every requirement is testable and has a verifying test in CL-DTT-001. Priority is **Must** unless stated.

3.1 Data intake (FR-IN)

ID	Requirement	Rationale	Test
FR-IN-01	The system shall store each activity record with: plant, scope, activity type, quantity, unit, activity start and end date, tier, source document reference, SHA-256 of the evidence file, entered-by, and a system-assigned recorded_at timestamp (UTC).	recorded_at is the record time that makes "as known at date X" possible.	TC-IN-01
FR-IN-02	Activity records shall be append-only . A correction is a new record with supersedes_id set. UPDATE and DELETE on activity records shall be rejected by the database.	History must not be rewritable; the audit trail is the data itself.	TC-IN-02
FR-IN-03	The tier shall be set from the evidence type: meter reading gives metered; supplier bill or invoice gives invoiced; any value derived from an assumption gives estimated. A user may downgrade a tier freely but may upgrade only by attaching evidence.	Prevents optimistic tiering, which would understate uncertainty.	TC-IN-03
FR-IN-04	A record whose interval crosses a month boundary shall be apportioned by calendar days into child records, each with the same tier and its own dates.	Each day must meet the factor valid on that day (FR-FAC-03).	TC-AW-11
FR-IN-05	For every plant, activity type and month the system shall hold an expected item . An expected item with no record at the time of check has the data source missing and becomes an Open Item.	"Missing" must be a recorded state, never silence or zero.	TC-DT-05
FR-IN-06	A supplier declaration shall be stored as evidence with supplier, coverage, value, unit, receipt date and file hash, and flagged "unverified". The declared value, when used, enters as tier estimated regardless of the supplier's claims.	Declarations vary in reliability; they inform but do not raise confidence.	TC-DT-04
FR-IN-07	A quantity of zero entered with evidence shall be a valid record. Absence of a record shall never be read as zero.	A shut plant and an unreported plant are different facts.	TC-AW-12

3.2 Calculation (FR-CALC)

ID	Requirement	Rationale	Test
FR-CALC-01	Emission of a record (kgCO ₂ e) = quantity, converted to the factor's unit, multiplied by the factor value valid on the record's activity date.	The calculation is deliberately simple so that audit effort goes to inputs.	TC-CA-01

3. Functional Requirements

ID	Requirement	Rationale	Test
FR-CALC-02	All arithmetic shall use decimal numbers with context precision 34 digits and rounding mode ROUND_HALF_EVEN. Each item emission is quantised to 6 decimal places once, after conversion and multiplication. Totals are exact sums of quantised items. Binary floating point shall not be used for any reported value.	Floating point can differ by platform and operation order; decimals cannot.	TC-CA-02
FR-CALC-03	Unit conversions shall come from the pinned reference release, not from code constants.	A conversion is reference data and must be versioned like a factor.	TC-CA-03
FR-CALC-04	The system shall compute totals by plant, by scope, by tier, and the grand total.	Needed for the customer report and for the per-plant data quality metrics.	TC-CA-04
FR-CALC-05	A record of kind DIRECT_EMISSION (a supplier-declared kgCO ₂ e figure) shall be used as given, with no factor, and shall be marked "no factor applied".	Suppliers often declare emissions, not quantities.	TC-CA-05
FR-CALC-06	At close, each Open Item shall be gap-filled with the quantity of the most recent earlier month, for the same plant and activity type, that is not itself gap-filled; tier estimated; flag GAP_FILLED. If no such basis exists, close shall be blocked.	Deterministic, explainable, and conservative in tier.	TC-AW-05

3.3 Emission factor versioning (FR-FAC)

ID	Requirement	Rationale	Test
FR-FAC-01	Emission factors shall be stored in reference releases . A sealed release is immutable: the database shall reject UPDATE, DELETE and INSERT on its rows.	Reproduction is possible by construction only if history cannot change.	TC-FV-01
FR-FAC-02	Each factor row shall carry authority, publication edition, activity type, unit, value, valid_from , valid_to (half-open interval [from, to) of activity dates), a row checksum, the checksum of the row it supersedes (if any), and loaded_at .	Valid time (when the factor applies) and record time (when we learned it) are kept apart.	TC-FV-02
FR-FAC-03	The factor for a record shall be the single row of the pinned release whose type and unit match and whose interval contains the record's activity date . If zero or more than one row matches, the calculation shall fail with an error. There shall be no fallback to a "latest" factor.	The factor valid at the time of the activity is the correct one; silent fallback is how history gets rewritten.	TC-FV-03
FR-FAC-04	Within a release, factor intervals for the same type and unit shall not overlap; this shall be enforced by a database exclusion constraint.	Makes FR-FAC-03 unambiguous by construction.	TC-FV-04

3. Functional Requirements

ID	Requirement	Rationale	Test
FR-FAC-05	A release shall have a manifest listing every row checksum, the tier percentages, unit conversions and rounding policy, and a SHA-256 manifest hash. The hash shall be re-verified when the release is loaded and when it is used.	Detects accidental or malicious change.	TC-FV-05
FR-FAC-06	A report shall record the Pair (calc release and reference release) it was computed with.	The pair, not the code alone, defines the number.	TC-RP-01
FR-FAC-07	Recomputing a report shall use its recorded Pair. A what-if run with newer factors shall be a separate mode that produces a document stamped COMPARISON, which cannot be issued or stored as a report.	Prevents "recomputation" from quietly giving a different answer.	TC-FV-06
FR-FAC-08	When the authority revises a factor retroactively, a new release shall be created; old releases and the reports pinned to them remain unchanged. A restatement may then be issued under FR-LATE-06.	Corrections are explicit events, not edits.	TC-FV-07

3.4 Uncertainty and its disclosure (FR-UNC)

Tier percentages are the project data: **metered 2%, invoiced 8%, estimated 25%**. They are read as relative uncertainty of the item's emission figure, at the same confidence level for all tiers. They are stored in the reference release so that a change to them is also versioned.

ID	Requirement	Rationale	Test
FR-UNC-01	The system shall assign each emission line the relative uncertainty of its tier: 2%, 8% or 25%.	Given project data.	TC-UN-01
FR-UNC-02	Item uncertainty $U(i)$ = tier percentage x emission $E(i)$, in kgCO _{2e} , quantised to 6 decimal places.	Absolute uncertainty is what adds up.	TC-UN-01
FR-UNC-03	For any aggregate (plant, scope, grand total) the system shall compute $U(rss)$ = square root of the sum of $U(i)$ squared and report it as the uncertainty of the aggregate. Square roots shall be computed in decimal arithmetic.	Independent errors partly cancel; root-sum-square is the standard method for adding independent uncertainties.	TC-UN-02
FR-UNC-04	The system shall also compute the worst-case bound $U(lin)$ = sum of $U(i)$, and disclose it beside $U(rss)$.	RSS assumes independence. A mis-calibrated meter or a shared assumption makes errors move together. Disclosing both brackets the truth instead of picking the flattering one.	TC-UN-03
FR-UNC-05	Every reported total shall be shown as $E \pm U$ (relative %) , with U rounded to 2 significant figures and E rounded to the same decimal place.	Prevents false precision: no digit is shown that the uncertainty does not support.	TC-UN-04
FR-UNC-06	Every report shall show the share of emissions by tier, per scope and in total, and the list of gap-filled and declared items.	A reader must see how much of the total rests on estimates.	TC-UN-05

3. Functional Requirements

ID	Requirement	Rationale	Test
FR-UNC-07	No screen, PDF or CSV shall show a total without its uncertainty.	Makes honesty the default rather than an option.	TC-UN-06
FR-UNC-08	Every report shall carry a fixed method statement: percentages used, RSS method, independence assumption, the worst-case bound, and the limitation that uncertainty of the emission factors is not included .	Limitations stated, not hidden.	TC-UN-07

3.5 Period close (FR-PC)

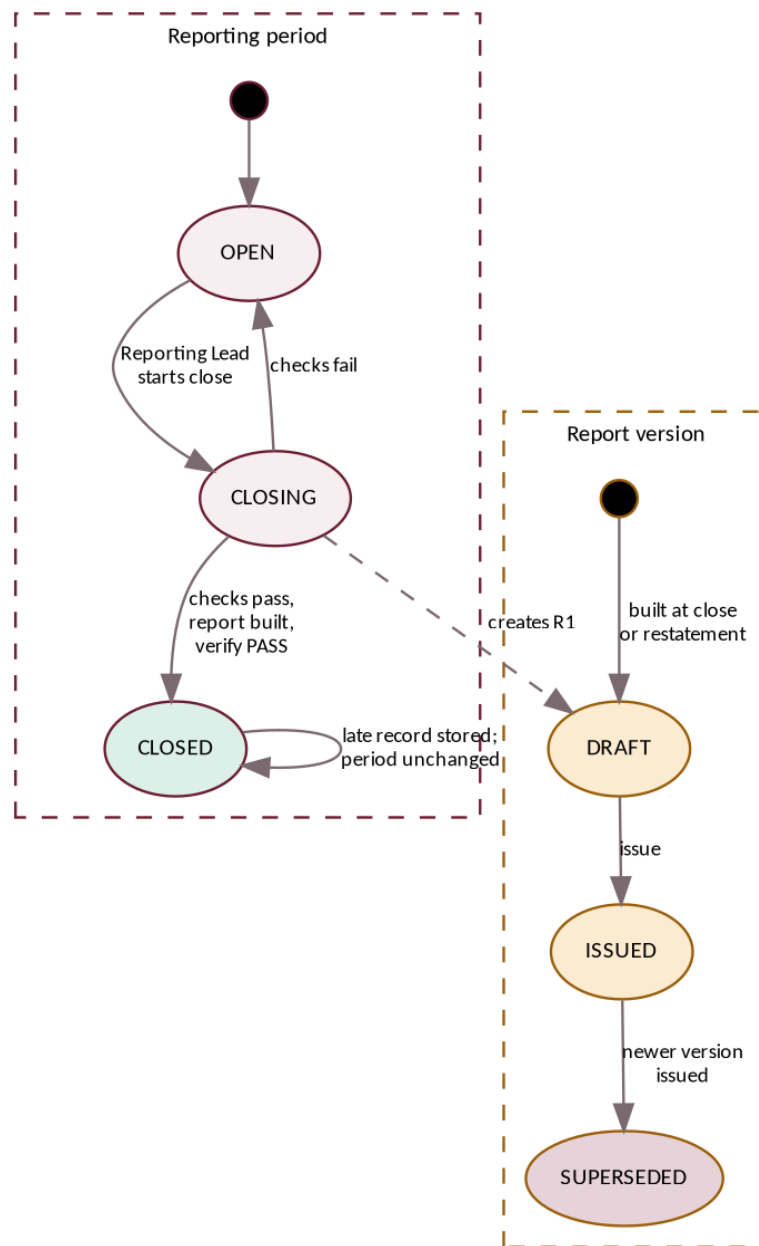


Figure 1. Period and report lifecycle.

3. Functional Requirements

ID	Requirement	Rationale	Test
FR-PC-01	A period shall be in exactly one state: OPEN, CLOSING or CLOSED. A report version shall be DRAFT, ISSUED or SUPERSEDED.	Explicit states give precise rules.	TC-PC-01
FR-PC-02	Close shall be refused unless: no Open Item remains unresolved; every metered and invoiced record has evidence; the Pair is valid and its hashes verify; the calculation completes with no factor-lookup error.	A report with known holes must not be issued as complete.	TC-PC-02
FR-PC-03	Close shall be performed only by the Reporting Lead. It shall freeze <code>closed_at</code> , pin the Pair, compute the report, store the canonical report and its hash, and run verification (FR-RPT-04).	Single, logged, atomic event.	TC-PC-03
FR-PC-04	The administrative data cut-off (set in the cycle calendar) is distinct from close. Records stored between the cut-off and close are accepted into the open period and flagged AFTER_CUTOFF for the timeliness metric. Only records stored after close are late.	Keeps the definition of "late" precise and testable.	TC-PC-04
FR-PC-05	The records included in a report shall be exactly those with <code>recorded_at</code> less than or equal to <code>closed_at</code> and not superseded as of that instant.	This single rule makes reproduction independent of later data.	TC-PC-05
FR-PC-06	A closed period shall never be reopened. Changes after close follow section 3.6.	A reopened period would invalidate the issued hash.	TC-PC-06

3.6 Late data after period close (FR-LATE)

Definition. A *late record* is any record, including an invoice, bill, meter reading or correction, whose activity date lies in a CLOSED period at the moment it is stored. What happens to it is defined step by step below and shown in Figure 2.

3. Functional Requirements

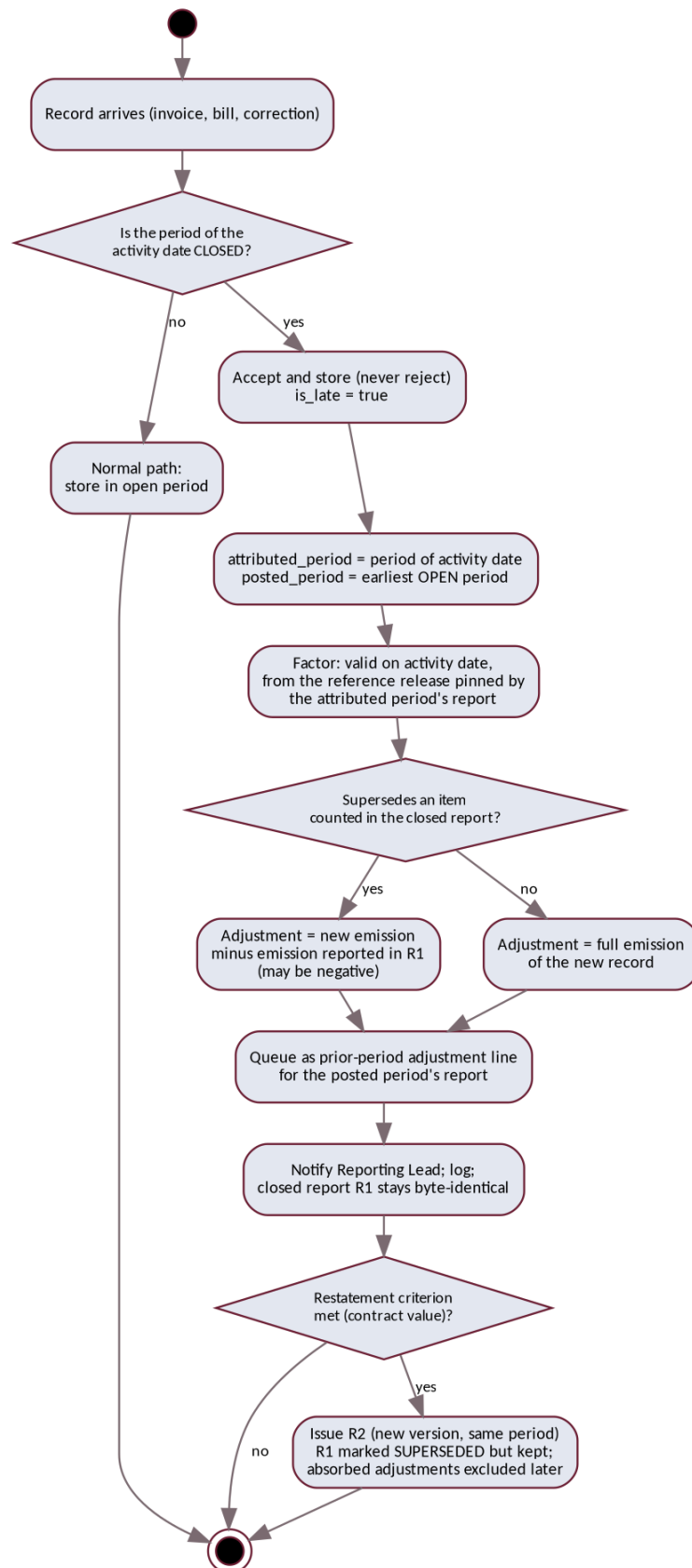


Figure 2. Processing of a late record.

3. Functional Requirements

ID	Requirement	Rationale	Test
FR-LATE-01	A late record shall be accepted and stored , flagged is_late , with attributed_period = the period of its activity date and posted_period = the earliest OPEN period at the time of storage. It shall never be rejected.	An invoice is a fact, whether or not the books are closed.	TC-LT-01
FR-LATE-02	The closed period's report, its inputs and its hash shall remain unchanged . Recomputing it shall still give identical bytes.	The issued figure is what the customer holds.	TC-LT-02
FR-LATE-03	The late record's emission shall use the factor valid on its activity date , taken from the reference release pinned by the attributed period's report .	Using the posting period's release or date would mix two factor sets inside one period's story.	TC-LT-03
FR-LATE-04	If the late record supersedes an item counted in the closed report (a gap-filled or estimated item, or a corrected value), the adjustment = new emission minus the emission most recently reported for that item (in the closed report, or in an earlier adjustment line if the item was already corrected once; it may be negative). Otherwise the adjustment is the full emission of the new record.	Only the difference is new information; the old amount is already reported.	TC-LT-04
FR-LATE-05	The adjustment shall appear in the report of the posted period as a separate prior-period adjustment line showing attributed period, amount, tier and uncertainty. That report shall show subtotals "this period" and "this period plus adjustments". Adjustments shall not be mixed silently into current-period totals.	The customer can reconcile both views.	TC-LT-05
FR-LATE-06	The Reporting Lead may restate a closed period when the adjustments attributed to it reach the restatement criterion defined by the customer contract (a configuration value; this SRS does not set it). Restatement creates a new report version for the same period (cut-off after the late data, new hash). The old version becomes SUPERSEDED but is kept and stays reproducible. Adjustments absorbed into the new version are marked ABSORBED and excluded from later reports.	Gives a defined path for material changes without editing history and without double counting.	TC-LT-06
FR-LATE-07	The Reporting Lead shall be notified of each late record with plant, period and size.	Visibility.	TC-LT-01
FR-LATE-08	If an item is still missing after close (it was gap-filled at close), the closed report keeps the gap-filled value and the item remains on the Late-Data Watchlist until data arrives. No calculation is repeated and no adjustment is posted until a record arrives.	Defines the "missing after close" case that spreadsheets leave vague.	TC-DT-11, TC-DT-12

3. Functional Requirements

ID	Requirement	Rationale	Test
FR-LATE-09	A supplier declaration that arrives after close for an item that was missing shall be treated as a late record of tier estimated with flag DECLARED (rule R10).	Same rules for all late evidence.	TC-DT-10

3.7 Reports and audit (FR-RPT, FR-AUD, FR-AC)

ID	Requirement	Rationale	Test
FR-RPT-01	A report shall contain: header (report id, version, period, plants, Pair, cut-off, input hash), totals by scope and plant with uncertainty, tier shares, gap-filled and declared items, prior-period adjustments, evidence index, method statement.	Everything an auditor needs on one document.	TC-RP-02
FR-RPT-02	The canonical report shall be JSON with keys sorted, no insignificant whitespace, UTF-8, LF line end, numbers as strings with fixed 6 decimals, dates as ISO 8601 (UTC), lines sorted by plant, scope, activity type, record id, and no run-time data (no run timestamp, host or user). The report hash is the SHA-256 of these bytes.	Same inputs must give the same bytes.	TC-RP-03
FR-RPT-03	PDF and CSV renderings shall be generated from the canonical JSON and shall print the report hash.	Renderings are derived, never a second source of truth.	TC-RP-02
FR-RPT-04	A verify function (<code>carbonledger verify REPORT_ID</code>) shall recompute the report from its Pair, input cut-off and stored inputs, compare the bytes with the stored canonical report, and return PASS or FAIL. The input hash shall be checked before calculation.	The system must be able to prove its own history.	TC-RP-04
FR-AUD-01	Ingest, close, restatement, release load and verify actions shall be written to an append-only audit log (who, what, when, before and after hashes).	Accountability.	TC-AU-01
FR-AUD-02	An Auditor shall be able to drill from any total to its lines, records, factor row and evidence file.	Auditability in practice.	TC-AU-02
FR-AC-01	Access shall follow the roles in 2.2. Plant Data Officers see only their plant. Only the Reference Data Steward can load a release; only the Reporting Lead can close or restate.	Separation of duties.	TC-AU-03

4. Non-Functional Requirements

ID	Requirement	Verification
NFR-REP-01	Reproducibility. For any issued report, verify shall return PASS, with byte-identical canonical output, at any later time while the Pair and inputs are retained, including after newer releases and newer calc modules exist.	TC-RP-04 to TC-RP-07
NFR-REP-02	Retention. Canonical reports, inputs snapshots, evidence files, sealed releases and the code for every Pair shall be retained for at least two years after issue, and longer if the contract requires.	Archive check, CL-CMP-001
NFR-REP-03	Determinism of environment. Calculations shall not depend on locale, time zone, system clock, hash ordering or file ordering.	TC-RP-05
NFR-AUD-01	Reference data and calculation changes shall follow the change control in CL-CMP-001 and shall not be possible by direct database edit.	Review and TC-FV-01
NFR-PERF-01	A full recompute of one period should complete in under 10 minutes on a standard laptop (design target for the expected volume).	TC-PF-01
NFR-SEC-01	Authentication for every user, roles enforced on the server, evidence files stored with their hash, HTTPS in deployment, daily backup tested by restore.	TC-AU-03
NFR-USE-01	A Plant Data Officer shall be able to enter a record with evidence in a single form, with units and tier chosen from lists.	Usability walkthrough in UAT
NFR-MNT-01	The calculation package shall be covered by the decision-table tests with 100% of the 12 rules and all 48 combinations exercised.	TC-DT-ALL

5 Design decisions and justifications

#	Decision	Alternatives considered	Why this one
D1	Factors keyed by activity date , not report date.	Use latest factor at report time.	The factor valid when the emission happened is the correct one; using the report date would change history each year.
D2	Sealed, immutable releases with copy-on-release rows.	Mutable factor table with history rows; a membership table linking releases to shared rows.	A native exclusion constraint per release enforces no overlap by construction. Data is small, so duplicating rows costs almost nothing.
D3	Reproduce as of a recorded_at cut-off over append-only records.	Snapshot copy of data for each report.	One rule (FR-PC-05) gives reproducibility without copying; the snapshot hash still proves the inputs.
D4	Pair = frozen calc module + release; old modules never edited.	Always run latest code.	If a defect is found in calc v1, v1 stays as it was so old reports still reproduce; the fix goes into v2 and is applied by explicit restatement.
D5	Decimal arithmetic with fixed rounding.	Floating point with tolerance comparison.	The requirement is byte equality, which tolerance cannot give.

5. Design decisions and justifications

#	Decision	Alternatives considered	Why this one
D6	Root-sum-square plus worst-case bound.	Linear sum only; RSS only.	RSS alone may flatter; linear alone overstates. Both disclosed.
D7	Late records never alter a closed report; adjustment goes to the next open period, restatement by rule.	Reopen the period; ignore late data.	Reopening breaks issued hashes; ignoring hides real emissions.
D8	Hash over canonical JSON, not PDF.	Hash the PDF.	PDF generators embed fonts and timestamps and are not byte-stable.
D9	Tier fixed by evidence; declaration never upgrades tier.	Trust the declaration as invoiced.	Declarations vary in reliability; conservative tiering is the honest choice.

6

Traceability

The Decision Table and Test Set (CL-DTT-001, section 9) maps every requirement ID above to its tests. The Design Pack (CL-DSG-001) maps each requirement group to a module and table. Reporting-cycle dates are in CL-PPL-001, and data quality targets in CL-DQP-001.